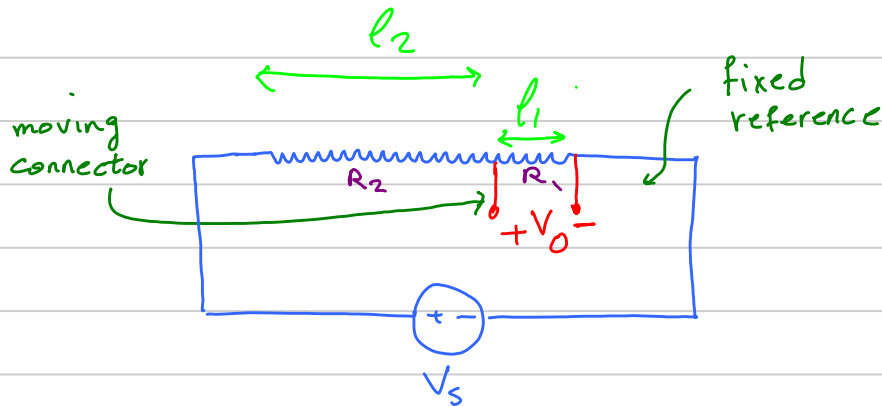




EX. Position sensor based on the voltage divider rule



voltage divider rule:

$$V_o = \frac{R_1}{R_1 + R_2} V_s, \quad R = \rho \frac{l}{A} \Rightarrow (R \propto l)$$

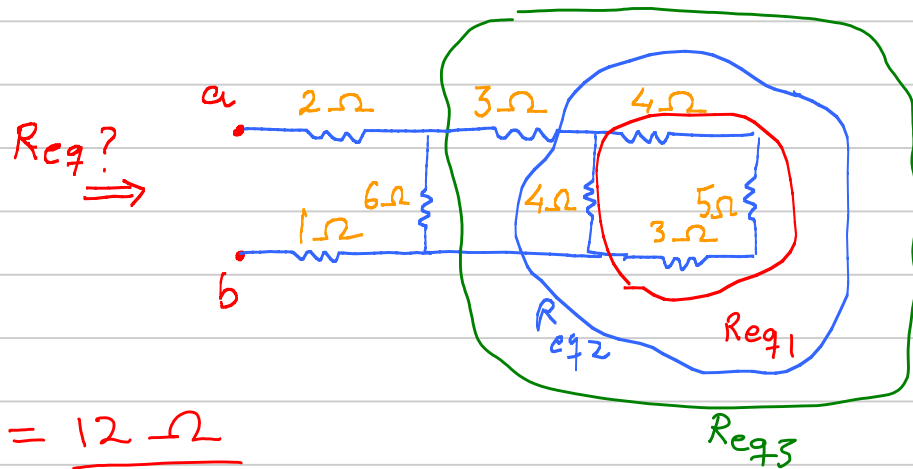
$$V_o = \frac{l_1}{l_1 + l_2} V_s = \frac{l_1}{L} V_s$$

← total length (constant)

$$\Rightarrow V_o \propto l_1$$

Therefore, the output voltage V_o is proportional to the distance of the connector from the reference. This can make a position sensor.

Ex. Combine the resistors and find the equivalent resistance.



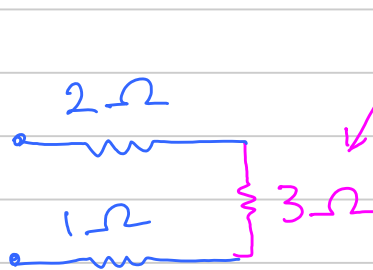
$$R_{eq1} = 4 + 5 + 3 = \underline{12\ \Omega}$$

$$R_{eq2} = R_{eq1} \parallel 4\ \Omega = 12 \parallel 4$$

$$= \frac{12 \times 4}{12 + 4} = \underline{3\ \Omega}$$

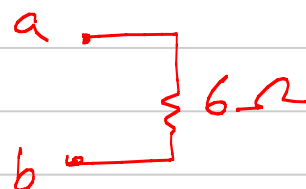
$$R_{eq3} = R_{eq2} + 3 = \underline{6\ \Omega}$$

$$R_{eq4} = R_{eq3} \parallel 6\ \Omega = 6 \parallel 6 = \underline{3\ \Omega}$$



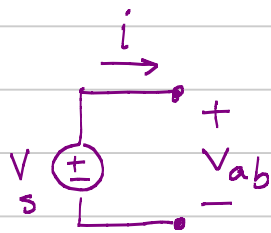
$$R_{eq} = 2 + R_{eq4} + 1$$

$$\boxed{R_{eq} = 6\ \Omega}$$



Practical Sources

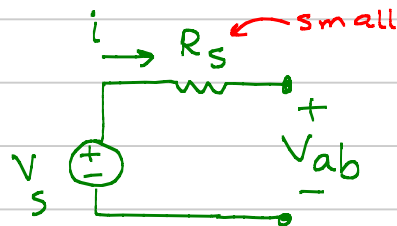
Practical sources have some internal resistances.



ideal voltage source

$$V_{ab} = V_s$$

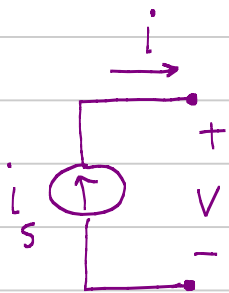
(independent of i)



practical voltage source

$$V_{ab} = V_s - R_s i$$

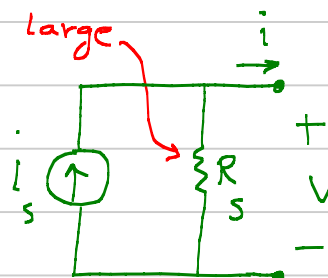
(V_{ab} depends on i , which is determined by the load)



ideal current source

$$i = i_s$$

(indep. of V)



practical current source

$$i = i_s - \frac{V}{R_s}$$

(depends on V)

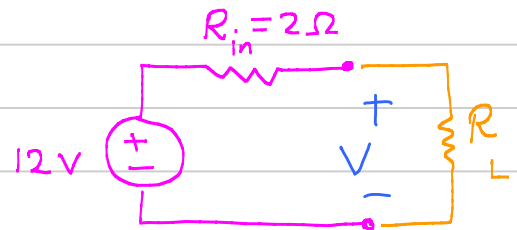
Ex. A 12V, 10 W light bulb is connected to a 12V battery with an internal resistance of 2 ohms. What is the voltage across the light bulb?

$$P = \frac{V^2}{R_L} \Rightarrow R_L = \frac{V^2}{P}$$

$$R_L = \frac{12^2}{10} = 14.4 \Omega$$

$$\text{VDR: } V = 12 \frac{R_L}{R_L + R_{in}} = \frac{14.4}{14.4 + 2}$$

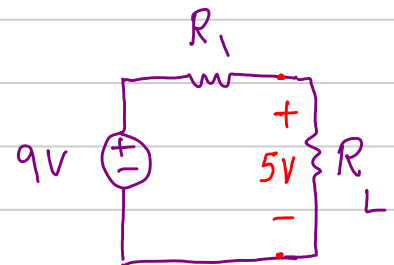
$$\underline{V = 10.5 \text{ V}}$$



Ex. Design an attenuator circuit to supply a 5V, 1W light bulb from a 9V battery.

First find the resistance of the bulb.

$$R_L = \frac{V^2}{P} = \frac{25}{1} = 25 \Omega$$



we need to apply only 5V across the bulb.

$$\text{VDR: } 5 = 9 \frac{R_L}{R_1 + R_L}$$

$$5 = 9 \frac{25}{R_1 + 25} \Rightarrow \underline{R_1 = 20 \Omega}$$

Ex: Find the power dissipated in the 6 ohm resistor.

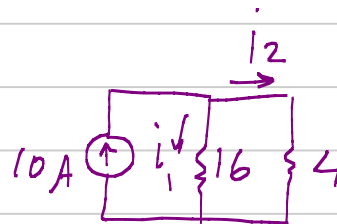
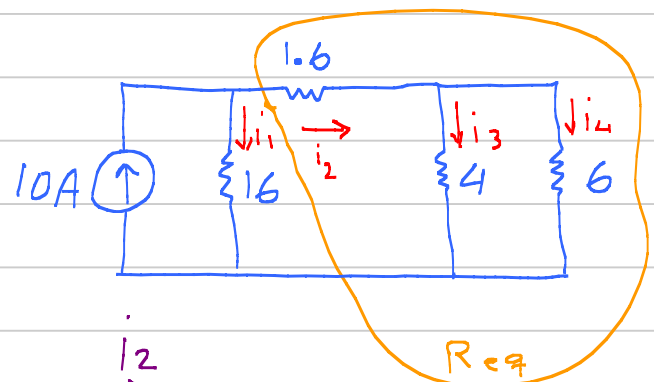
$$R_{eq} = (4 \parallel 6) + 1.6$$

$$= \frac{4 \times 6}{4 + 6} + 1.6 = 4 \Omega$$

$$i_2 = 10 \frac{16}{4 + 16} = 8 A$$

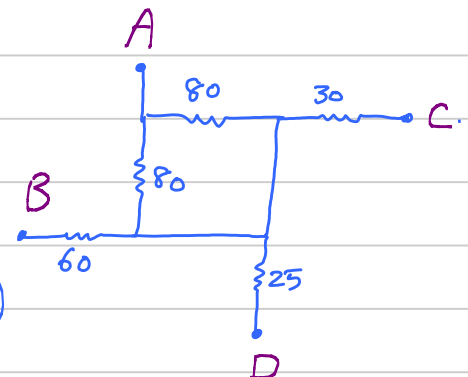
$$i_4 = i_2 \frac{4}{4 + 6} = 8 \frac{4}{10} = 3.2 A$$

$$P_{6\Omega} = R i_4^2 = 6 (3.2)^2 = \underline{61.44 W}$$



The equivalent circuit of a network is different at different terminals.

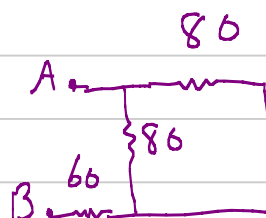
Ex. Find The equivalent resistance looking into terminals AB, AC, BC, and BD.



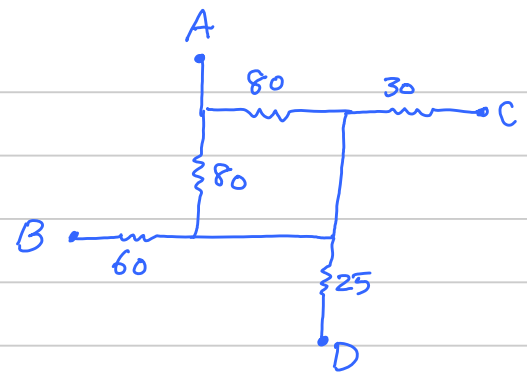
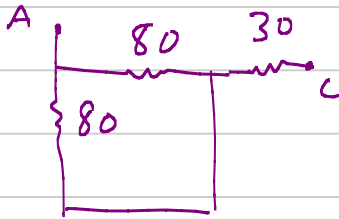
R_{AB} : (When C and D are disconnected)

$$R_{AB} = (80 \parallel 80) + 60$$

$$R_{AB} = 100 \Omega$$



$R_{AC} :$

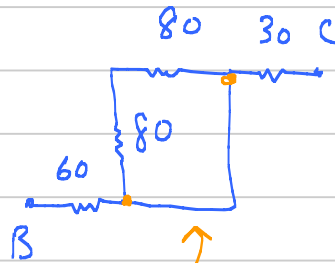


$$R_{AC} = (80 \parallel 80) + 30 = 70 \Omega$$

$R_{BC} :$

$$(80 + 80) \parallel 0 = 0$$

↑
short ckt



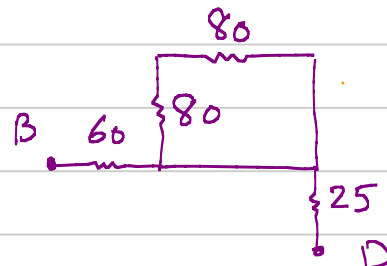
$$R_{BC} = 60 + 0 + 30 = 90 \Omega$$

A resistor in parallel with a short ckt can be removed.

R_{BD} (when A & C are open):

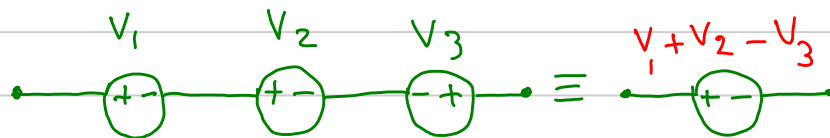
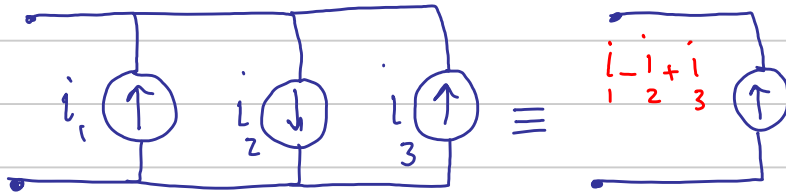
$$(80 + 80) \parallel 0 = 0$$

$$R_{BD} = 60 + 0 + 25 = 85 \Omega$$



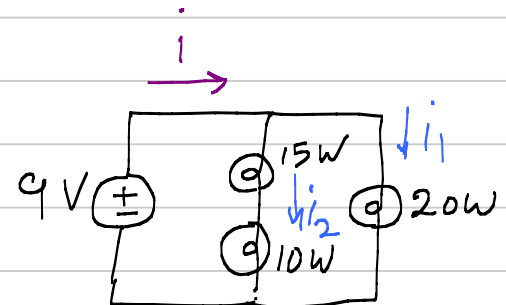
Combining Sources

We can combine parallel connected current sources and series connected voltage sources.



Ex. Three light bulbs are connected to a 9v battery. find

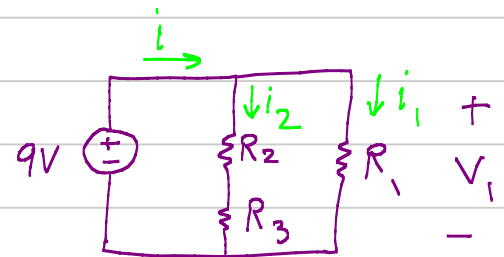
- total current supplied by the battery.
- the current in each light bulb.
- the resistance of each bulb.



$$a) \sum P = 0 \Rightarrow P_{9V} + 15 + 10 + 20 = 0$$

$$P_{9V} = -45W$$

$$-i = \frac{P}{V} \Rightarrow i = \frac{45}{9} = \underline{5A}$$



$$b) 20 = 9 \times i_1 \Rightarrow i_1 = \frac{20}{9} = \underline{2.22A}$$

$$i_2 = i - i_1 = 5 - 2.22 = \underline{2.78A}$$

$$R_1 = \frac{9}{2.22} = 4.05\Omega, \quad R_2 = \frac{P}{i_2^2} = 1.94\Omega, \quad R_3 = \frac{10}{2.78^2} = 1.3\Omega$$